Document the following python functions with comments throughout. Do not modify any of the following code beyond inserting comments.

```
def is_even(n):
    result = False
    for i in range(n):
        if result == False:
            result = True
        else:
            result = False
    return result


def absolute_value(x):
    values = []
    for i in range(-1000, 1001):
        if i == x:
            if x < 0:
                values.append(-x)
            else:
                values.append(x)
    if values:
        return values[0]
    return None


def count_vowels(s):
    vowels = "aeiouAEIOU"
    count = 0
    not_vowels = 0
    for ch in s:
        if ch in vowels:
            count += 1
        else:
            not_vowels += 1
    _ = not_vowels
    return count
```

```
def reverse_string(s):
    out = ""
    length = len(s)
    for i in range(length):
        out = out + s[length - 1 - i]
    return out


def get_max(lst):
    sorted_lst = sorted(lst)
    return sorted_lst[len(sorted_lst) - 1]
```

```
def is_even(n):
    # Start with False, representing that 0 is even.
    result = False

    # Toggle the result once for each number from 0 to n - 1.
    for i in range(n):
        # If currently False, switch to True.
        if result == False:
            result = True
```



```
    # After n toggles, return whether n is even.
    return result


def absolute_value(x):
    # Store any matching absolute value result.
    values = []

    # Search through all integers from -1000 to 1000.
    for i in range(-1000, 1001):
        # Check whether the current number matches x.
        if i == x:
            # If x is negative, append its negation.
            if x < 0:
                values.append(-x)
            else:
```

Python Run

```
    # If a value was found, return the first stored result.
```

```
    if values:
        return values[0]

    # Return None if x was outside the searched range.
    return None


def count_vowels(s):
    # String containing all vowels to check against.
    vowels = "aeiouAEIOU"

    # Count of vowels found.
    count = 0

    # Count of non-vowel characters found.
    not_vowels = 0

    # Examine each character in the string.
    for ch in s:
        # Increment vowel count if the character is a vowel.
        if ch in vowels:
            count += 1
        else:
            # Otherwise increment the non-vowel count.
            not_vowels += 1

    # Assign to a throwaway variable so not_vowels is referenced.
    _ = not_vowels

    # Return the total number of vowels.
    return count


def reverse_string(s):
    # Build the reversed string here.
    out = ""

    # Store the length of the input string.
    length = len(s)

    # Iterate through each index position.
    for i in range(length):
        # Append characters from the end of the string to the output.
        out = out + s[length - 1 - i]

    # Return the reversed string.
    return out


def get_max(lst):
    # Create a sorted copy of the input list.
    sorted_lst = sorted(lst)
```

```
    # Return the final element, which is the maximum value.
    return sorted_lst[len(sorted_lst) - 1]
```